At least three of $\beta(2), \beta(4), \ldots, \beta(364)$ are irrational.

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Abstract

We prove that at least three of the numbers $\beta(2), \beta(4), \ldots, \beta(364)$ are irrational, where β is the Dirichlet beta function.

1 Introduction

For $s > 1$, the Dirichlet beta function is

$$\beta(s) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^s} = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \cdots .$$

Theorem 1.1. *At least three of $\beta(2), \beta(4), \ldots, \beta(364)$ are irrational.*

1.1 Comparison with earlier results

Rivoal and Zudilin [11] proved that one of $\beta(2), \beta(4), \ldots, \beta(14)$ is irrational. The endpoint was reduced to 12 by Zudilin [14] and to 10 by Lai and Zhou [8]. OpenAI [10, Theorem 1.1] subsequently proved that Catalan's constant $\beta(2)$ is irrational. Krisman [4] proved that at least one of $\beta(4), \beta(6), \ldots, \beta(36)$ is irrational.

Sorokin [12, Theorem S] proved that, for every sufficiently large odd b and every even $a > 100b \log b$, at least one of

$$\beta(b+1), \beta(b+3), \ldots, \beta(b+a-1)$$

is irrational. Huttner [3, Remark 5] obtains the same conclusion. Both results have a computable threshold for b, but neither gives its numerical value. To our knowledge, the explicit count of three irrational values through 364 in Theorem 1.1 is new.

Asymptotic results give bounds valid for all sufficiently large endpoints. In [13], we improve the previous logarithmic lower bounds by proving that, for all sufficiently large even s, at least $0.21\sqrt{s/\log s}$ of $\beta(2), \beta(4), \ldots, \beta(s)$ are linearly independent over the rationals. The proofs of asymptotic results can also yield bounds at specified finite endpoints, so their consequences at $s = 364$ must be checked. Appendix A makes this comparison for the finite estimates of Rivoal–Zudilin, Nishimoto and Fischler, and for the bound in [13]. The estimates and parameter families examined there do not yield the three-value count in Theorem 1.1.

1.2 Canceling two possible exceptions

Suppose, for a contradiction, that at most two of the listed beta values are irrational. We construct three sequences of linear forms. A fixed integer combination of the three forms at each index cancels the possible irrational terms, producing a single sequence of rational numbers. Clearing denominators yields a sequence of integers tending to zero with infinitely many nonzero terms. This is impossible: a nonzero integer has absolute value at least 1.

Three sequences allow us to cancel two coefficients. At each positive even index n, the forms have the shape

$$r_{n,b} = q_b(n) + \sum_{\substack{2\le i\le 364\\ i \text{ even}}} b^i \rho_i(n)\beta(i), \qquad b = 1, 2, 3, \tag{1.1}$$

with rational coefficients. The coefficient column at $\beta(i)$ is a scalar multiple of the fixed vector $(1, 2^i, 3^i)^{\mathsf{T}}$. A nonzero integer vector perpendicular to two such vectors gives weights that cancel the same two beta terms for every n.

For example, the combination

$$H_n = 15r_{n,1} - 6r_{n,2} + r_{n,3}$$

has coefficient $\rho_i(n)(15 - 6 \cdot 2^i + 3^i)$ at $\beta(i)$. This vanishes for $i = 2$ and $i = 4$.

For any chosen pair, fix such weights and write $H_n = \sum_{b=1}^{3} w_b r_{n,b}$, with constant term $q_w(n) = \sum_{b=1}^{3} w_b q_b(n)$. Sections 3 and 4 give a common denominator for the coefficients and show that multiplying by it preserves decay. The remaining beta values are rational under our assumption. Multiplying by their fixed common denominator then gives the required integers tending to zero.

We must still show that H_n is nonzero infinitely often. The constant term could cancel the remaining beta terms, so we distinguish them by reduction modulo a large prime p. Section 5 compares the forms at $n = 2p$ with those at $n = 2$. After multiplication by a suitable power of p, every remaining beta term vanishes modulo p, while the constant term is congruent to a nonzero multiple of $q_w(2)$. Thus $q_w(2) \neq 0$ implies $H_{2p} \neq 0$ for all sufficiently large p. Section 6 verifies $q_w(2) \neq 0$ for every chosen pair by an exact determinant calculation. Section 7 combines these facts to finish the contradiction.

The construction builds on the rational-function and elimination methods of Kumar and Tosi [5] and Lai and Yu [7]. Lai and Sprang [6] also use prime dilation to prove nonvanishing. Our proof uses partial fractions and elementary prime valuations, together with Stirling's formula, the prime number theorem in its least-common-multiple form, and Wilson's and Fermat's congruences.

2 Three forms with matching coefficient columns

2.1 Designing one function for all three grids

To obtain the coefficient pattern in (1.1), we evaluate a single rational function R_n on each grid j/b, for $b = 1, 2, 3$, and sum with the alternating signs and odd indices of the beta series.

Write these signs as weights $\chi_4(j)$: $\chi_4(2k) = 0$ and $\chi_4(2k+1) = (-1)^k$ for integers $k \ge 0$. Thus the weights are $1, 0, -1, 0, 1, 0, -1, 0, \ldots$ for $j = 1, 2, \ldots$, and

$$\sum_{j=1}^{\infty} \chi_4(j) R_n(j/b) = R_n(1/b) - R_n(3/b) + R_n(5/b) - \cdots.$$

The weights have period 4. We therefore space the poles by 4: a partial-fraction term at $-4h$ satisfies

$$\frac{1}{(j/b + 4h)^i} = \frac{b^i}{(j + 4bh)^i}.$$

Since $4bh$ is a multiple of 4, the shift preserves the weights. Poles at $0, -4, \ldots, -4n$ therefore give the required factor b^i.

The numerator serves two purposes. Reflection about $-2n$ pairs the poles; we arrange that $R_n(-4n - t) = R_n(t)$ for even n, so the paired coefficients cancel at every odd order.

We also make the initial terms of each alternating sum vanish, leaving only a tail. If the last positive zero is rn, its reflection is $-(r+4)n$. The blocks of zeros must therefore span an interval of length $(2r+4)n$.

The three grids have fractional parts $0, 1/2, 1/3, 2/3$. We use the representatives

$$\Theta = \{1, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{2}{3}\},$$

where 1 represents the integer class. We choose $r = 7$, giving a block length of $18n$, and take the largest beta index to be 364. Longer zero blocks remove more initial terms but increase the numerator degree. Higher pole orders improve decay but enlarge both the list of beta values and the multiplier needed to clear denominators. Sections 4 and 6 verify that our choices give sufficient decay and the required nonvanishing.

For positive even n, the numerator with these zero blocks is

$$P_n(t) = (t - 7n) \prod_{\theta \in \Theta} \prod_{j=0}^{18n-1} (t - 7n + j + \theta). \tag{2.1}$$

The factor $t - 7n$ adds the right endpoint to the integer block. The integer roots of P_n are exactly $-11n, -11n+1, \ldots, 7n$; the half-integer block reflects into itself, and the third-shift blocks exchange. In particular, P_n has a simple zero at each proposed pole $-4h$. To leave pole order 364 after cancellation, we use the denominator $\prod_{h=0}^{n}(t+4h)^{365}$.

2.2 Choosing the arithmetic normalization

Multiplication by a constant preserves the zeros, poles and reflection symmetry. We choose this constant to control the denominators of the partial-fraction coefficients. For each numerator block, we divide by $(n!)^{18}$, using the divisibility of products of n consecutive terms. There are four blocks, hence the denominator $(n!)^{72}$. The fractional shifts require compensating powers of 2 and 3, supplied by K^n with $K = 2^{36}3^{54}$. Section 3 proves the resulting divisibility bounds.

Expansion at a pole $-4h$ introduces further denominators. The constant terms of the other pole factors have product

$$\prod_{\substack{0 \le k \le n \\ k \ne h}} (4(k-h))^{365} = 4^{365n}(-1)^{365h}\big(h!(n-h)!\big)^{365}.$$

Multiplication by 4^{365n} cancels the power of 4. Multiplication by $((n+1)!)^{365}$ then leaves the integer factor $(-1)^{365h}((n+1)\binom{n}{h})^{365}$. We therefore set

$$C_n = 4^{365n} K^n \frac{((n+1)!)^{365}}{(n!)^{72}}, \qquad R_n(t) = \frac{C_n P_n(t)}{\prod_{h=0}^{n}(t+4h)^{365}}. \tag{2.2}$$

The size of C_n must be included in the decay estimate of Section 4. The numerator has degree $72n+1$, so R_n has degree at infinity $-293n - 364$. Each pole has order exactly 364. Write

$$R_n(t) = \sum_{h=0}^{n} \sum_{i=1}^{364} \frac{a_{i,h}(n)}{(t+4h)^i}, \qquad \rho_i(n) = \sum_{h=0}^{n} a_{i,h}(n). \tag{2.3}$$

Lemma 2.1. *For $b = 1, 2, 3$, define*

$$r_{n,b} = \sum_{j=1}^{\infty} \chi_4(j) R_n(j/b), \tag{2.4}$$

$$q_b(n) = -\sum_{h=0}^{n} \sum_{i=1}^{364} a_{i,h}(n) b^i \sum_{j=1}^{4bh} \frac{\chi_4(j)}{j^i}. \tag{2.5}$$

Then

$$r_{n,b} = q_b(n) + \sum_{\substack{2 \le i \le 364 \\ i \text{ even}}} b^i \rho_i(n) \beta(i). \tag{2.6}$$

Moreover,

$$|r_{n,b}| \le \sum_{j > 7bn} R_n(j/b). \tag{2.7}$$

Proof. The roots of P_n are symmetric about $-2n$. The integer and half-integer blocks each reflect into themselves, and the one-third and two-thirds blocks exchange. Because n is even, numerator and denominator change by the same sign under $t \mapsto -4n - t$. Therefore

$$R_n(-4n - t) = R_n(t), \qquad a_{i,n-h}(n) = (-1)^i a_{i,h}(n), \qquad \rho_i(n) = 0 \quad (i \text{ odd}).$$

For each partial fraction, shifting the summation variable by $4bh$ preserves χ_4 and gives

$$\sum_{j=1}^{\infty} \frac{\chi_4(j)}{(j/b + 4h)^i} = b^i \left(\beta(i) - \sum_{j=1}^{4bh} \frac{\chi_4(j)}{j^i} \right).$$

Here $\beta(1)$ denotes its convergent alternating series. The $i = 1$ sums converge because the partial sums of the weights are bounded; there are only finitely many partial-fraction sums. The sum of the $\beta(1)$ coefficients is zero, as is the sum at every other odd index. This proves (2.6). The original series (2.4) is absolutely convergent.

Every positive grid point $j/b \le 7n$ is a numerator zero. Beyond $7n$, all factors of R_n are positive. Taking absolute values now gives (2.7). We use this positive tail only to bound size; nonvanishing of the alternating sum requires a separate argument. □

3 A common denominator

Put $d_n = \operatorname{lcm}(1, \ldots, n)$ and $\delta_n = 4d_n$. For a prime p, $v_p(x)$ is the exponent of p in a nonzero rational number x, with $v_p(0) = +\infty$. A rational number is *integral at p* if its valuation is nonnegative. We bound the denominators of both the beta coefficients and the rational correction terms.

Proposition 3.1. *For $1 \le i \le 364$, $0 \le h \le n$ and $b = 1, 2, 3$,*

$$\delta_n^{365-i} a_{i,h}(n) \in \mathbb{Z}, \qquad \delta_n^{365-i} \rho_i(n) \in \mathbb{Z}, \qquad \delta_n^{365} q_b(n) \in \mathbb{Z}. \tag{3.1}$$

3.1 Taylor coefficients of the numerator

Define

$$F_\theta(t) = \frac{A_\theta^n}{(n!)^{18}} \prod_{j=0}^{18n-1} (t - 7n + j + \theta), \qquad A_1 = 1, \quad A_{1/2} = 2^{36}, \quad A_{1/3} = A_{2/3} = 3^{27}.$$

For every integer u and every $k \geq 0$, we claim

$$d_n^k [t^k] F_\theta(t+u) \in \mathbb{Z}. \tag{3.2}$$

Here $[t^k]$ denotes the coefficient of t^k. Write $\theta = a/b$ in lowest terms. Multiplying each linear factor of $F_\theta(t+u)$ by b gives integer constant terms in a progression with step b. Fix a prime $q \nmid b$ and split this progression into 18 blocks of length n. For every $q^v \leq n$, at least $18\lfloor n/q^v \rfloor$ entries are divisible by q^v. To form a term in the coefficient of t^k, we choose the linear term from k factors and constant terms from the rest. Omitting those k constants loses at most k contributions at each prime-power level. Before division by $(n!)^{18}$, the resulting valuation is at least

$$18 v_q(n!) - k\lfloor \log_q n \rfloor.$$

Multiplication by d_n^k compensates for this loss. Any term that still contains a zero constant is zero and needs no estimate.

The factor A_θ^n compensates for primes dividing b. For $b = 2$, after clearing the $18n$ factor denominators, 2^{18n} remains; this covers $18v_2(n!) \leq 18n$. For $b = 3$, 3^{9n} remains, covering $18v_3(n!) \leq 9n$. Coefficient extraction introduces only further integer factors. Thus (3.2) holds at every prime.

Let

$$F(t) = (t - 7n) \prod_{\theta \in \Theta} F_\theta(t) = \frac{K^n P_n(t)}{(n!)^{72}}.$$

Expansion at $-4h$ gives

$$\begin{aligned} a_{i,h}(n) = (n+1)^{365} (-1)^{365h} \binom{n}{h}^{365} \\ \times [t^{365-i}] \left\{ F(t-4h) \prod_{\substack{0 \leq k \leq n \\ k \neq h}} \left(1 + \frac{t}{4(k-h)}\right)^{-365} \right\}. \end{aligned} \tag{3.3}$$

Each $4(k-h)$ divides δ_n. The binomial series for the remaining factors, together with (3.2), proves the first assertion of (3.1); summing over h proves the second. The cancellation of the constant powers of 4 uses the factor 4^{365n} in C_n; the same factor also contributes to the decay estimate in Section 4.

3.2 Clearing the rational correction terms

The denominators j in (2.5) can exceed n, so multiplication by δ_n^{365} need not make each summand integral. We group the terms with the same h, j by summing over the pole orders i. At any fixed prime, at most one pole group can fail to be integral after scaling. At a zero of R_n, the sum of all pole groups is zero. The possible exceptional group is therefore the negative of a sum of integral groups, and is integral as well.

Fix an odd j with $1 \leq j \leq 4bh$. The point $t_0 = j/b - 4h$ lies in $(-4n, 0)$ and is a numerator zero: its reduced denominator divides b, and Θ supplies the required fraction class. It is not a pole, since $t_0 = -4k$ would imply $j = 4b(h-k)$, impossible for odd j. Evaluating $R_n(t_0) = 0$ yields

$$\sum_i \frac{a_{i,h}(n)}{(j/b)^i} = -\sum_{k \neq h} \sum_i \frac{a_{i,k}(n)}{(j/b + 4(k-h))^i}. \tag{3.4}$$

Fix a prime q, put $v = v_q(\delta_n)$, and set $x_k = j/b + 4(k-h)$. At most one x_k can have $v_q(x_k) > v$: two such numbers would have a difference with valuation greater than v, whereas

$$x_k - x_l = 4(k-l) \mid \delta_n \qquad (k \neq l).$$

The coefficient bound already proved implies, for every nonzero rational x,

$$v_q\left(\frac{\delta_n^{365}a_{i,h}(n)}{x^i}\right) \geq i\big(v - v_q(x)\big). \tag{3.5}$$

If $v_q(x_h) \leq v$, this inequality makes every term on the left of (3.4) integral at q after multiplication by δ_n^{365}. If $v_q(x_h) > v$, every other x_k satisfies $v_q(x_k) \leq v$. The scaled right side is then integral at q, and the identity proves the same for the left side. Thus every scaled group is integral at every prime. Summing the groups in (2.5) proves the last assertion of Proposition 3.1.

4 Decay survives denominator clearing

The prime number theorem gives $d_n^{1/n} \to \mathrm{e}$, hence also $\delta_n^{1/n} \to \mathrm{e}$. The multiplier δ_n^{365} therefore grows at exponential rate e^{365}. The following bound shows that the forms decay fast enough to compensate.

Proposition 4.1. *For $b = 1, 2, 3$, along positive even n,*

$$\limsup_{n\to\infty} |\delta_n^{365} r_{n,b}|^{1/n} < \frac{3}{4}.$$

Proof. The positive tail begins just beyond the last zero $7n$. For fixed $x > 0$, its logarithmic growth rate at distance nx from that zero is

$$\Phi(x) := \lim_{\substack{n\to\infty \\ n \text{ even}}} \frac{1}{n} \log R_n\big(n(7+x)\big).$$

Thus, for fixed $x > 0$, $R_n(7n + nx) = \exp(n\Phi(x) + o(n))$. We first find the maximum of Φ, then use uniform estimates to bound the whole tail.

The product identity $\prod_{j=0}^{N-1}(z+j) = \Gamma(z+N)/\Gamma(z)$ gives

$$P_n(7n+t) = t \prod_{\theta\in\Theta} \frac{\Gamma(t+18n+\theta)}{\Gamma(t+\theta)},$$

$$\prod_{h=0}^{n} (7n+t+4h) = 4^{n+1} \frac{\Gamma((11n+t)/4+1)}{\Gamma((7n+t)/4)}.$$

Substitute these expressions and C_n from (2.2) into $\log R_n(7n+t)$, set $t = nx$, and apply Stirling's formula. The limit exists and equals

$$\begin{aligned}\Phi(x) = {} & 36\log 2 + 54\log 3 + 4\big((x+18)\log(x+18) - x\log x\big) \\ & - \frac{365}{4}\big((x+11)\log(x+11) - (x+7)\log(x+7)\big) + 365\log 4,\end{aligned} \tag{4.1}$$

with continuous extension to $x = 0$ obtained by setting $x\log x = 0$. The $n\log n$ terms and the linear terms from Stirling's formula cancel. The factor 4^{365n} in C_n contributes the final term $365\log 4$.

To locate the maximum, put

$$4\Phi'(x) = \log f(x), \qquad f(x) = \left(\frac{x+18}{x}\right)^{16} \left(\frac{x+7}{x+11}\right)^{365}. \tag{4.2}$$

Its logarithmic derivative is

$$(\log f)'(x) = -\frac{288}{x(x+18)} + \frac{1460}{(x+7)(x+11)}.$$

The numerator over a common positive denominator is $1172x^2 + 21096x - 22176$, which has one positive root. Thus $\log f$ first decreases and then increases. It tends to $+\infty$ as $x \to 0$ and to zero from below as $x \to \infty$, with leading term $-1172/x$. Consequently $f(x) = 1$ has exactly one positive solution x_0, where Φ attains its global maximum. Using $\Phi'(x_0) = 0$ to simplify (4.1) gives

$$G := \mathrm{e}^{\Phi(x_0)} = 2^{36}3^{54}(x_0+18)^{72}4^{365}\left(\frac{(x_0+7)^7}{(x_0+11)^{11}}\right)^{365/4}. \tag{4.3}$$

We now bound the sum in (2.7). For fixed A, the gamma-product formula gives

$$\log R_n(7n+t) = n\Phi(t/n) + O_A(\log n) \quad (1/3 \le t \le An),$$

uniformly on that interval. The estimate remains valid for small t because $\log\Gamma(y) = y\log y - y + O(\log(y+1)+1)$ for y bounded below by a fixed positive constant. Each grid has $O_A(n)$ terms in this interval, so the sum over the interval has nth-root limsup at most G.

For $t \ge An$ with $A \ge 18$, every numerator factor is at most $2t$ and every denominator factor is at least t. Therefore

$$R_n(7n+t) \le C_n 2^{72n+1} t^{-M_n}, \qquad M_n = 293n + 364.$$

The grid has spacing $1/b$. For the sum of t^{-M_n} over this grid, we bound the first term by $(An)^{-M_n}$ and the rest by an integral:

$$\begin{aligned}\sum_{j/b \ge (7+A)n} R_n(j/b) &\le C_n 2^{72n+1}\left((An)^{-M_n} + b\int_{An}^{\infty} t^{-M_n}\,dt\right)\\ &= C_n 2^{72n+1}(An)^{-M_n}\left(1 + \frac{bAn}{M_n - 1}\right).\end{aligned}$$

Stirling's formula gives

$$C_n = 4^{365n}K^n(n+1)^{365}(n!)^{293} = \left(4^{365}K\mathrm{e}^{-293}\right)^n n^{293n}\exp\bigl(O(\log n)\bigr).$$

The factor n^{293n} cancels the same power in $(An)^{-M_n}$. The remaining powers of n do not affect the nth-root limit. Taking nth roots in the tail bound therefore gives a limsup of at most

$$4^{365}K2^{72}\mathrm{e}^{-293}A^{-293}.$$

This tends to zero as $A \to \infty$. We conclude that

$$\limsup_{\substack{n\to\infty \\ n \text{ even}}} |r_{n,b}|^{1/n} \le G. \tag{4.4}$$

To prove $\mathrm{e}^{365}G < 3/4$, set $u = 3/5000$. Exact rational arithmetic gives

$$\left(\frac{u+18}{u}\right)^{16}\left(\frac{u+7}{u+11}\right)^{365} < 1, \tag{4.5}$$

so $x_0 < u$. Also $(x+7)^7/(x+11)^{11}$ is decreasing for $x \ge 0$, since its logarithmic derivative is $-4x/((x+7)(x+11))$. Finally, the positive exponential series gives

$$\mathrm{e} < \sum_{k=0}^{5}\frac{1}{k!} + \frac{1/6!}{1-1/7} = \frac{163}{60} + \frac{7}{4320} < \frac{68}{25}.$$

Consequently

$$(\mathrm{e}^{365}G)^4 < \left(\frac{68}{25}\right)^{1460} 2^{144}3^{216}\left(18+\frac{3}{5000}\right)^{288} 4^{1460}\left(\frac{7^7}{11^{11}}\right)^{365}$$
$$< \frac{81}{256} = \left(\frac{3}{4}\right)^4. \tag{4.6}$$

The rational expression in the first line is approximately 0.2761910348; its exact comparison with 81/256 is verified below. Combining (4.4) with $\delta_n^{1/n} \to \mathrm{e}$ proves the proposition. □

The comparisons (4.5) and (4.6), including the bound for e, can be reproduced with the following exact calculation:

```
from fractions import Fraction as F
u = F(3, 5000)
assert ((u+18)/u)**16 * ((u+7)/(u+11))**365 < 1
assert F(163,60) + F(7,4320) < F(68,25)
U = (F(68,25)**1460 * 2**144 * 3**216 * (18+u)**288
     * 4**1460 * (F(7)**7/F(11)**11)**365)
assert U < F(81,256)
```

5 Prime dilation and nonvanishing

The estimates above do not rule out $H_n = 0$: the constant term could cancel the remaining beta terms. We rule this out at indices $n = mp$, where m is a fixed positive even integer and p is a large prime. After multiplication by p^{364}, the beta coefficients vanish modulo p, while the constant terms reduce to a common nonzero multiple of their values at m. Under the rationality assumption, the remaining beta terms also vanish modulo all sufficiently large p. Hence a nonzero combined constant at m will imply $H_{mp} \neq 0$. We will take $m = 2$.

Write $\mathbb{Z}_{(p)}$ for the rational numbers whose denominators are prime to p. A *unit* in this ring has both numerator and denominator prime to p, and hence a nonzero reduction modulo p. All reductions below are of quantities proved to be p-integral.

5.1 How a long product reproduces a shorter one

For example, take $p = 5$ and consider $\prod_{j=1}^{10}(5z+j)$. The factors with $j = 5, 10$ are $5(z+1)$ and $5(z+2)$. Removing their two powers of 5 leaves

$$5^{-2}\prod_{j=1}^{10}(5z+j) = (z+1)(z+2)\prod_{\substack{1\le j\le 10\\ 5\nmid j}}(5z+j).$$

The remaining factors reduce to the constants $1, 2, 3, 4$, each occurring twice. Since $1\cdot 2\cdot 3\cdot 4 \equiv -1 \pmod 5$, their product is 1 modulo 5. Thus selecting every fifth factor recovers the product of length two:

$$5^{-2}\prod_{j=1}^{10}(5z+j) \equiv (z+1)(z+2) \pmod 5.$$

Fractional shifts can change which block is recovered. For example, in $5z+j+1/3$, divisibility by 5 means $j \equiv 3 \pmod 5$. Writing $j = 5v+3$ gives

$$5z + (5v+3) + \tfrac{1}{3} = 5\big(z+v+\tfrac{2}{3}\big).$$

The shift $1/3$ has become $2/3$; the companion block with shift $2/3$ returns $1/3$. The two blocks exchange, so their product is preserved.

We apply this selection to every numerator and denominator block of $R_{mp}(pz)$. The factors divisible by p recover the blocks of R_m, and Wilson's theorem evaluates the products of the remaining factors. Accounting for the powers of p and the normalization gives the following proposition.

Proposition 5.1. *For fixed positive even m and all sufficiently large primes p,*

$$p^{364}\rho_i(mp) \in p^i\mathbb{Z}_{(p)} \quad (1 \le i \le 364), \tag{5.1}$$

$$p^{364}q_b(mp) \equiv \chi_4(p)(m+1)^{-365}q_b(m) \pmod{p} \quad (b = 1, 2, 3). \tag{5.2}$$

5.2 The rational-function congruence

We first prove

$$p^{364}R_{mp}(pz) \equiv (m+1)^{-365}R_m(z) \pmod{p}. \tag{5.3}$$

This is an identity of rational functions over $\mathbb{F}_p$.

For $p > m+1$, both $(mp)!$ and $(mp+1)!$ have valuation m. After division by p^m, their units are $(-1)^m m! = m!$ modulo p, by Wilson's theorem and the evenness of m. The explicit powers of 2, 3 and 4 in the normalization reduce to those in C_m by Fermat's theorem. Thus C_{mp} has valuation $293m$, and its unit reduces to $C_m/(m+1)^{365}$.

The extra factor $pz - 7mp$ in the numerator contributes one factor p. In a block with shift $\theta = a/b$, precisely $18m$ factors are divisible by p. They take the form

$$p(z - 7m + v + \theta'), \qquad 0 \le v < 18m, \qquad \theta' = a'/b, \quad 1 \le a' \le b, \quad pa' \equiv a \pmod{b}.$$

The map $\theta \mapsto \theta'$ permutes Θ. When $p \equiv 2 \pmod 3$, the one-third and two-thirds blocks exchange, leaving their product unchanged. After division by p, the selected factors reproduce $P_m(z)$. In each block the remaining factors have product $(-1)^{18m} = 1$ modulo p.

The divisible denominator factors are $p(z+4y)$, $0 \le y \le m$, each with multiplicity 365. The remaining denominator unit is $(-1)^{365m} = 1$. The net exponent of p is therefore

$$293m + (72m + 1) - 365(m+1) = -364.$$

The units and remaining factors give exactly (5.3).

5.3 Reducing the partial-fraction coefficients

We extract the principal parts from (5.3) by expanding at each pole. At a selected pole $z = -4y$, put $z = -4y + u$. The factorization above shows that

$$u^{365}p^{364}R_{mp}(p(u-4y)) \in \mathbb{Z}_{(p)}[[u]]. \tag{5.4}$$

Multiplication by u^{365} removes the denominator factor at that pole. After extraction of the powers of p already counted, every other denominator factor has a unit constant term. Its reciprocal has p-integral power-series coefficients. Reduction therefore commutes with coefficient extraction, giving in $\mathbb{F}_p[[u]]$

$$u^{365}p^{364}R_{mp}(p(u-4y)) \equiv (m+1)^{-365}u^{365}R_m(u-4y) \pmod{p}.$$

The term $a_{i,py}(mp)/(pu)^i$ contributes $p^{364-i}a_{i,py}(mp)$ to the coefficient of u^{365-i} on the left. Terms regular at this pole contribute only from degree 365 onward. Comparing coefficients yields

$$p^{364-i}a_{i,py}(mp) \equiv (m+1)^{-365}a_{i,y}(m) \pmod{p}. \tag{5.5}$$

We must also control the poles not indexed by a multiple of p. Write $h = x + py$ with $1 \le x \le p-1$ and substitute $t = pz - 4x$. The extra numerator factor is now a unit. Each of the four numerator blocks has $18m$ divisible factors, while the denominator has m divisible factors, each with multiplicity 365. The total valuation is

$$293m + 72m - 365m = 0.$$

After removing these powers of p, the remaining poles $-4y$ are distinct modulo p. The same power-series argument, now applied to $u^{365}R_{mp}(p(u-4y)-4x)$, shows that

$$p^{-i}a_{i,h}(mp) \in \mathbb{Z}_{(p)} \qquad (p \nmid h). \tag{5.6}$$

Both (5.5) and (5.6) imply $p^{364}a_{i,h}(mp) \in p^i\mathbb{Z}_{(p)}$; summing over h proves (5.1).

5.4 Reducing the constant terms

Apply these coefficient bounds to the formula (2.5) for $q_b(mp)$. If $p > 12m$, every summation index $j \le 4bmp$ is smaller than p^2, so it has valuation at most one. After multiplication by p^{364}, the terms with $p \nmid h$ vanish modulo p by (5.6), even when $p \mid j$. For $h = py$, terms with $p \nmid j$ vanish by (5.5). Only terms with both $p \mid h$ and $p \mid j$ can remain. Write

$$h = py, \qquad j = pk, \qquad 1 \le k \le 4by.$$

The coefficients reduce by (5.5). For odd p, the weights satisfy $\chi_4(pk) = \chi_4(p)\chi_4(k)$: multiplication by p preserves the two odd residue classes modulo 4 if $p \equiv 1 \pmod 4$ and exchanges them if $p \equiv 3 \pmod 4$; even k has weight zero. Taking $\chi_4(p)$ outside the reduced finite sum gives (5.2) and completes the proof of Proposition 5.1.

6 The finite certificate at $m = 2$

Fix even indices $2 \le i < j \le 364$, and write

$$v_k = (1, 2^k, 3^k)^{\mathsf{T}}, \qquad q(2) = (q_1(2), q_2(2), q_3(2))^{\mathsf{T}}.$$

The integer vector $w = v_i \times v_j$ is perpendicular to both columns, so the combination $\sum_{b=1}^{3} w_b r_{n,b}$ cancels $\beta(i)$ and $\beta(j)$ for every n. Its constant term at $n = 2$ is

$$q_w(2) = w \cdot q(2) = \det\big(q(2), v_i, v_j\big).$$

The determinant is the constant left after canceling the two beta terms. If it is nonzero, Proposition 5.1 proves nonvanishing at $n = 2p$ for all sufficiently large p under the rationality assumption. We therefore need the following finite statement.

Lemma 6.1 (Exact seed certificate)**.** *For every pair of distinct even indices $2 \le i < j \le 364$,*

$$\det \begin{pmatrix} q_1(2) & 1 & 1 \\ q_2(2) & 2^i & 2^j \\ q_3(2) & 3^i & 3^j \end{pmatrix} \neq 0. \tag{6.1}$$

We prove this by reduction modulo the prime $\ell = 1\,000\,003$. A nonzero residue proves that the rational determinant is nonzero, provided its entries have denominators prime to ℓ. We first give the three seed residues and test all pairs. Their derivation follows in Subsection 6.2.

6.1 The determinant test

The common normalization $C_2 = 2^{1825}3^{473}$ is nonzero modulo ℓ, so we may compute with

$$U_2(t) = \frac{P_2(t)}{t^{365}(t+4)^{365}(t+8)^{365}}.$$

Let $\widetilde{a}_{i,h}$ be the partial-fraction coefficients of U_2 and $\widetilde{q}_b$ the corresponding constant terms, defined as in (2.3) and (2.5). Then $q_b(2) = C_2\widetilde{q}_b$. The denominators of $\widetilde{a}_{i,h}$ have only the prime factors 2 and 3; the finite sums add only primes at most 23. Reduction modulo ℓ is therefore legitimate.

The calculation in Subsection 6.2 gives

$$(\widetilde{q}_1, \widetilde{q}_2, \widetilde{q}_3) \equiv (614660, 346158, 44336) \pmod{\ell}. \tag{6.2}$$

For $Q = (614660, 346158, 44336)$, compute, modulo ℓ,

$$D_{i,j} = Q_1(2^i3^j - 2^j3^i) - Q_2(3^j - 3^i) + Q_3(2^j - 2^i), \qquad 2 \le i < j \le 364, \quad i, j \text{ even}.$$

The following loop checks all pairs using the residues in (6.2):

```
ell = 1000003
Q1, Q2, Q3 = 614660, 346158, 44336
count = 0
for i in range(2, 365, 2):
    for j in range(i + 2, 365, 2):
        det = (Q1 * (pow(2,i,ell)*pow(3,j,ell)
                     - pow(2,j,ell)*pow(3,i,ell))
               - Q2 * (pow(3,j,ell)-pow(3,i,ell))
               + Q3 * (pow(2,j,ell)-pow(2,i,ell))) % ell
        assert det != 0
        count += 1
assert count == 16471
```

All $\binom{182}{2} = 16471$ residues are nonzero. Since C_2 is also nonzero modulo ℓ, this proves (6.1) over $\mathbb{Q}$. The weights $(15, -6, 1)$ from the introduction give

$$15 \cdot 614660 - 6 \cdot 346158 + 44336 \equiv 187267 \not\equiv 0 \pmod{\ell}. \tag{6.3}$$

6.2 Recovering the seed residues

To obtain (6.2), we compute the partial fractions of U_2. At a pole $-4h$, substitute $t = u - 4h$ and multiply by u^{364}. The resulting function is regular at $u = 0$. Its first 364 Taylor coefficients give the principal part of U_2, in reverse order.

At the pole $-4h$, where $h = 0, 1, 2$, form the multiset

$$\mathcal{C}_h = \{-4h - 14\} \cup \{-4h - 14 + j + \theta : 0 \le j < 36,\ \theta \in \Theta\}.$$

It has exactly one zero. Delete that zero to obtain $\mathcal{C}_h^*$, and put $\mathcal{D}_h = \{4(k-h) : 0 \le k \le 2,\ k \ne h\}$. Thus

$$B_h(u) := u^{364}U_2(u - 4h) = \frac{\prod_{c\in\mathcal{C}_h^*}(u+c)}{\prod_{d\in\mathcal{D}_h}(u+d)^{365}} = \sum_{j\ge 0}\gamma_{h,j}u^j.$$

Its logarithmic derivative gives a convenient exact recurrence:

$$\begin{aligned}
\gamma_{h,0} &= \frac{\prod_{c\in\mathcal{C}_h^*} c}{\prod_{d\in\mathcal{D}_h} d^{365}},\\
T_{h,k} &= \sum_{c\in\mathcal{C}_h^*} c^{-k} - 365\sum_{d\in\mathcal{D}_h} d^{-k},\\
\gamma_{h,j} &= \frac{1}{j}\sum_{k=1}^{j}(-1)^{k-1}T_{h,k}\gamma_{h,j-k} \quad (1\le j\le 363),\\
\widetilde{a}_{i,h} &= \gamma_{h,364-i} \quad (1\le i\le 364).
\end{aligned} \tag{6.4}$$

All divisions are valid modulo ℓ. Each nonzero $c\in\mathcal{C}_h^*$ and $d\in\mathcal{D}_h$ has numerator and denominator of absolute value less than ℓ, and each recurrence index j is at most 363.

Substitution in (2.5) gives the residues in (6.2). The supplied implementations in `checks/seed.py` and `checks/verify.py` use only Python's standard library. Primality of ℓ is checked by trial division up to its integer square root.

The checker computes all 1092 partial-fraction coefficients both by (6.4) and by multiplying truncated power series. The results agree in every entry. It also checks rational-function reconstruction at three points and the vanishing of all odd ρ_i modulo ℓ. A separate calculation over $\mathbb{Q}$ verifies the divisibility assertions of Proposition 3.1 at $n=2$ for all 1092 coefficients and all three constant terms, and recovers (6.2).

7 The irrationality contradiction

Proof of Theorem 1.1. Suppose at most two of $\beta(2),\beta(4),\dots,\beta(364)$ are irrational. Choose distinct even indices i,j in $[2,364]$ such that every other value in the list is rational. The columns $v_i=(1,2^i,3^i)^{\mathsf{T}}$ and v_j are independent. Their cross product gives a nonzero integer vector $w=(w_1,w_2,w_3)$ with $w\cdot v_i=w\cdot v_j=0$. By Lemma 6.1,

$$q_w(2) := \sum_{b=1}^{3} w_b q_b(2) \neq 0. \tag{7.1}$$

Fix this vector for the rest of the argument and put $H_n=\sum_{b=1}^{3} w_b r_{n,b}$.

Equation (2.6) expresses H_n as a rational constant plus a rational linear combination of the remaining beta values. All those values are rational by assumption. Let D be a positive common denominator of these finitely many rational values. Proposition 3.1 shows that

$$D\delta_n^{365}H_n\in\mathbb{Z}. \tag{7.2}$$

Because the weights are fixed, Proposition 4.1 makes these integers tend to zero exponentially along even n.

It remains to show that infinitely many are nonzero. Take $n=2p$, where p is a sufficiently large prime, coprime to D and to the numerator and denominator of $q_w(2)$. By (5.2),

$$p^{364}\sum_{b=1}^{3} w_b q_b(2p) \equiv \chi_4(p)3^{-365}q_w(2) \not\equiv 0 \pmod{p}.$$

For each remaining index k, the rational number $\beta(k)$ is p-integral because $p\nmid D$. After the same scaling, its coefficient is divisible by p^k by (5.1) and the integrality of the weights. Thus every remaining beta term reduces to zero modulo p, while the constant term does not. Hence $H_{2p}\neq 0$.

The integers in (7.2) are therefore nonzero for infinitely many even n, although they tend to zero. This contradiction proves the theorem. □

7.1 Reproduction and review status

Run `python3 checks/verify.py` from the paper directory. It uses only the Python standard library and writes `checks/verification.json`. The finite seed, all elimination pairs, the two decay comparisons and the normalized seed denominators are checked exactly. It also tests the dilation identity at selected primes in all four residue classes modulo 12; these sampled evaluations are consistency checks, whereas Section 5 proves the general congruence.

An independent proof audit supplied on 3 November 2026 reported no substantive mathematical gap in the preceding plain-text proof and independently reconstructed its finite certificate without using the supplied verification scripts. This manuscript includes the auditor's requested local-power-series explanation and reorganizes the exposition. The audit applies to the recorded plain-text version; this typeset revision has not received a new independent audit. The review record identifies the exact version and scope. Neither the computations nor the reported audit constitute formal verification.

A Finite consequences of asymptotic estimates

To decide whether an asymptotic result already gives Theorem 1.1, we examine its finite-parameter estimates. Write

$$\delta(s) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{1, \beta(2), \beta(4), \dots, \beta(s)\} \qquad (s \text{ even}).$$

A bound forcing $\delta(364) \geq 4$ would imply three irrational values. The comparisons below concern the specified estimates and parameter families in the cited proofs. They give either a dimension bound too small to force 4, even after integer rounding, or a size bound that does not establish decay of the integer linear forms.

A.1 Rivoal–Zudilin's uniform family

For the parameters in [11, (26)], put $q = 366$ and let $r = 2, 4, \dots, 182$. Equation (29) of that paper gives

$$\delta(q-2) \geq \frac{q\log r + (q-2r-1)\log 2}{(2+\log 2)(q-1) + (2r+1)\log(2r+1)}.$$

The largest right-hand side is approximately 1.18156, at $r = 28$. The sharper estimate preceding it also remains below 2. Indeed, evaluate the maximum defining its Laplace factor at $t_1 = \cdots = t_q = r/(r+1)$ and use $(1 + t_1 \cdots t_q)^{2r+1} \leq 2^{2r+1}$. This bounds that sharper dimension estimate *above* by

$$U(q,r) = \frac{q\{\log 2 + (r+1)\log(r+1) - r\log r\}}{(2+\log 2)(q-1) + (2r+1)\log(2r+1)} < 2. \tag{A.1}$$

For all 91 choices of r, the final inequality follows from the exact integer comparison

$$8^{4(q-1)} 2^{q-2} (2r+1)^{2(2r+1)} r^{qr} > 3^{4(q-1)} (r+1)^{q(r+1)}$$

and $\mathrm{e} > 8/3$. Also $U(q,r)$ increases with q for fixed $r \geq 2$, so the same conclusion holds at smaller endpoints in this family. These estimates can certify $\delta(364) \geq 2$, hence one irrational value, but cannot force dimension 4.

A.2 Nishimoto's explicit estimate

In [9, Theorem 5.5], take the periodic coefficients to be χ_4, with period d a positive multiple of 4. For positive integers a, b with a even, put $r = (d + 2b)/d$ and $\Delta = (r + 1)\log(r + 1) - (r - 1)\log(r - 1)$. The explicit estimate is $\delta(a) \geq 1 + \mathcal{A}/\mathcal{B}$, where

$$\mathcal{A} = (a + d)\Delta - 2a(d + \log 2) - 4b(\log d - \log 2) - 2dr \log(2r) - \tfrac{1}{3},$$
$$\mathcal{B} = 2ad + 2a \log 2 + 4(b + d)\log(b + d) - 4d \log d.$$

Among the hypotheses are $r \geq 2$ and $(a + d)/d \geq 3r$. For $a \leq 364$ these imply $a \geq 5d$ and $r + 1 < 32$, so

$$\Delta < 2\log(r + 1) + 2 < 10 \log 2 + 2 < 9.$$

Using $\log 2 > 1/2$ and discarding negative terms gives $\mathcal{A} < (8 - 2d)a + 9d$. For $d = 4$, therefore, $\mathcal{A} < 36$ and $\mathcal{B} > 9a \geq 180$, whence $1 + \mathcal{A}/\mathcal{B} < 6/5$. For $d \geq 8$, we instead obtain $\mathcal{A} < d(49 - 10d) < 0$. Thus this explicit estimate cannot force dimension 4 at 364. For comparison, the period-4 entries in [9, Table 5] certify dimension at least 2 at 2594 and at least 3 at 990206.

A.3 Fischler's size estimates

The estimates in [1, Section 4.2 and Proposition 4.4] and [2, Proposition 1, (1.1)] bound the integer linear forms by $\alpha^{n+o(n)}$. Their applications require $\alpha < 1$. With the subscripts distinguishing the two papers, the explicit factors are

$$\alpha_{18} = \mathrm{e}^a 4^{a/N - r}(N + 1)^{2r+2} r^{-a/N+4r+2}, \qquad 1 \leq r < \frac{a}{2N},$$
$$\alpha_{20} = (4\mathrm{e})^{(a+1)/N}(2N)^{2r+2} r^{-(a+1)/N+4(r+1)}, \qquad 1 \leq r < \frac{a}{3N}.$$

Here a, N, r are positive integers. For even beta values we may take $a = 365$, since the largest even index is then 364. In the first construction, the ordinary representation of χ_4 uses $4 \mid N$, while its twisted representation permits $N \equiv 2 \pmod 4$, including $N = 2$. We therefore check all positive even N. In the second construction we check all positive multiples of 4, including the enlarged periods used in the irrational-value count.

There are 429 admissible pairs (N, r) in the first case and 111 in the second. Every pair gives $\alpha > 1$. These are exact checks: raise α to the power N, replace e by $8/3$, and compare the resulting positive integers after clearing denominators. For fixed N, r, each $\log \alpha$ is affine in a and positive at $a = 0$. Positivity at $a = 365$ therefore also covers smaller endpoints. Thus these size estimates do not establish the decay needed at 364.

A.4 The square-root bound

For our result in [13], we check the finite dimension estimate before the asymptotic choice of parameters. The argument below covers all admissible choices at this endpoint. In that paper's notation, put $d = a + h$ and $H_d = \sum_{j=1}^{d} 1/j$. The largest beta index is $2\lfloor d/2 \rfloor$, so an endpoint at most 364 means $d \leq 365$. Let X be the coefficient-growth exponent $\log \chi$ in that construction. Equations (2.2), (5.2) and (5.3) of [13] give

$$X \geq \frac{\Omega^2 \log r}{2(a - \omega)},$$
$$\log \alpha = X - \Omega \log r + \kappa(1 + \log 4 + H_d),$$
$$\log B = X + (r + 1)\log(3/2) + \kappa(\log 8 + H_d).$$

Here a, h are integers with $h \geq 0$, $r \geq 2$, and $a > \Omega \geq \omega > \kappa > 2r$. The dimension estimate is $\delta(2\lfloor d/2 \rfloor) \geq 1 + R$, where $R = -\log \alpha / \log B$ when $\alpha < 1$. We show that $R < 2$, so this estimate cannot force dimension 4.

If $R \geq 2$, rearranging the last two formulas yields

$$\Omega \log r - 3X \geq \kappa(1 + \log 4 + 2 \log 8 + 3H_d) + 2(r+1)\log(3/2).$$

The lower bound on X makes the left side at most the maximum of a quadratic in Ω, namely $(a-\omega)\log r/6$. Since $\kappa > 2r$ and $\log r/r \leq 1/\mathrm{e}$, division by κ gives

$$\frac{\Omega \log r - 3X}{\kappa} < \frac{365}{12\mathrm{e}} < \frac{365}{32}.$$

On the other hand, $d \geq 5$, $\log 4 > 1$ and $\log 8 > 2$ make the right side, divided by κ, greater than

$$6 + 3H_5 = \frac{257}{20} > \frac{365}{32},$$

a contradiction.

The accompanying script `checks/check_asymptotic_comparisons.py` reproduces the finite integer comparisons and the rational constants used above. It uses only Python's standard library; its decimal summaries are illustrative and are not used to decide any inequality.

References

[1] S. Fischler, *Shidlovsky's multiplicity estimate and irrationality of zeta values*, J. Aust. Math. Soc. **105** (2018), no. 2, 145–172. doi:10.1017/S1446788717000386.

[2] S. Fischler, *Irrationality of values of L-functions of Dirichlet characters*, J. Lond. Math. Soc. **101** (2020), no. 2, 857–876. doi:10.1112/jlms.12290. We use the author revision of 10 October 2019.

[3] M. Huttner, *Riemann P-scheme, monodromy and diophantine approximations*, Indag. Math. **23** (2012), 522–546. doi:10.1016/j.indag.2012.03.002.

[4] S. Krisman, *One of $\beta(4), \beta(6), \ldots, \beta(36)$ is irrational*, preprint, 2026. Hexagon 2610.00285v2.

[5] V. Kumar and R. Tosi, *Irrationality of values of L-functions for real Dirichlet characters*, preprint, 30 October 2026. arXiv:2610.41458v1.

[6] L. Lai and J. Sprang, *Many p-adic odd zeta values are irrational*, preprint, version of 17 February 2025. arXiv:2306.10393v2.

[7] L. Lai and P. Yu, *A note on the number of irrational odd zeta values*, Compos. Math. **156** (2020), no. 8, 1699–1717. arXiv:1911.08458v2.

[8] L. Lai and L. Zhou, *At least two of $\zeta(5), \zeta(7), \ldots, \zeta(35)$ are irrational*, arXiv:2103.00904v2, Section 6, Theorem 6.1.

[9] M. Nishimoto, *On the linear independence of the special values of a Dirichlet series with periodic coefficients*, 2011. arXiv:1102.3247v1.

[10] OpenAI, *Catalan's constant is irrational*, preprint, 24 September 2026. OpenAI Math.

[11] T. Rivoal and W. Zudilin, *Diophantine properties of numbers related to Catalan's constant*, Math. Ann. **326** (2003), 705–721. doi:10.1007/s00208-003-0420-2.

[12] V. N. Sorokin, *On the Zudilin–Rivoal theorem*, Mat. Zametki **81** (2007), no. 6, 912–923; English translation: Math. Notes **81** (2007), no. 6, 817–826. doi:10.4213/mzm3742.

[13] twinheterodox, *Linear independence of even Dirichlet beta values*, 2026. Author version of 3 November 2026.

[14] W. Zudilin, *Arithmetic of Catalan's constant and its relatives*, Abh. Math. Semin. Univ. Hambg. **89** (2019), 45–53. arXiv:1804.09922v2.